Reg. No. | | | | | | | | | | |

[22MSCH3H01]

THIRD SEMESTER UNIVERSITY EXAMINATION, NOVEMBER 2023

SPECTROSCOPY–II

Time: 03 HOURS

Maximum Marks: 50

SECTION – A

1. **Answer any FOUR of the following questions.** 4×5=20

   a. Describe the significance of $A_{\parallel}$ and $g_{\parallel}$ to analyze the metal complex.

   b. What is the intrinsic lifetime of a fluorophore? Derive the expression of intrinsic lifetime.

   c. What is degeneracy? How does it affect the rotational energy intensity profiles?

   d. Discuss the S electron density on the isomeric shift.

   e. Show that the asymmetric stretch of H_2O molecule belongs to B_2 representation.

SECTION – B

2. **Answer any TWO of the following questions.** 2×9=18

   a. What is FRET? Explain with an energy diagram. Describe the factors guiding FRET.

   b. Explain the difference in isomeric shift between the ^{57}Fe and ^{119}Sn.

   c. Explain "symmetry elements" and "symmetry operations" with suitable examples.

SECTION – C

3. **Answer the following question.** 12

   a. i. Give the difference between ESR and NMR spectroscopic techniques and mention the limitation of ESR.

      ii. Draw the spectral line in Hydrogen under the influence of an external electric field.

      iii. Explain the Mulliken symbols B_{2g} and A_{1u}.